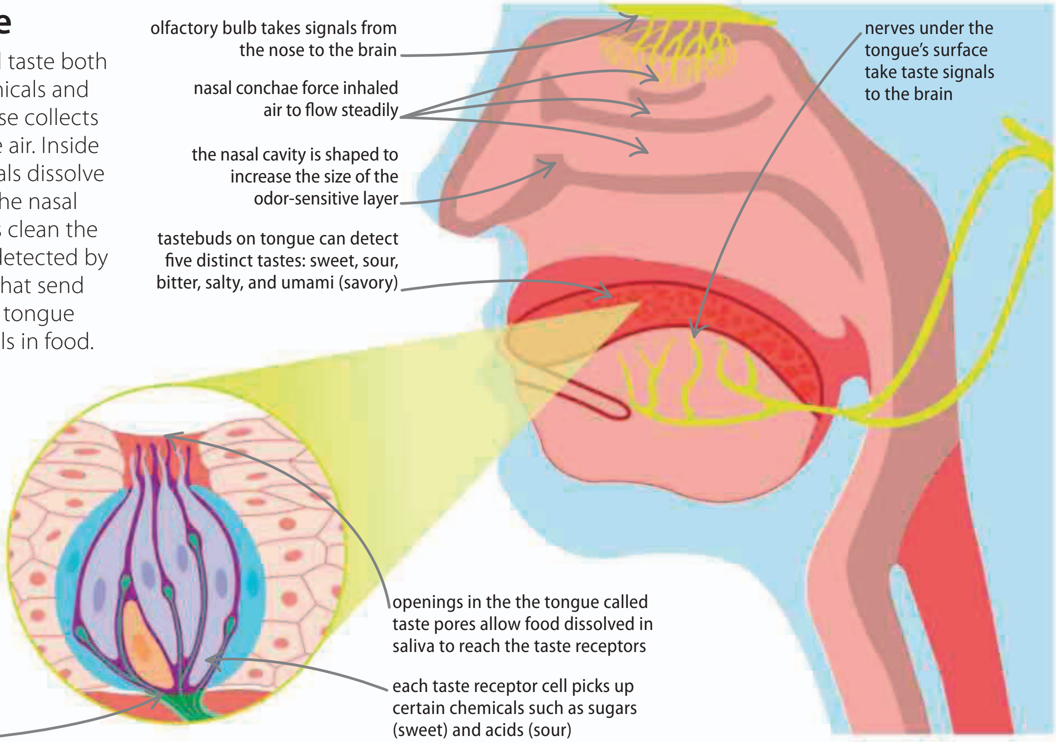
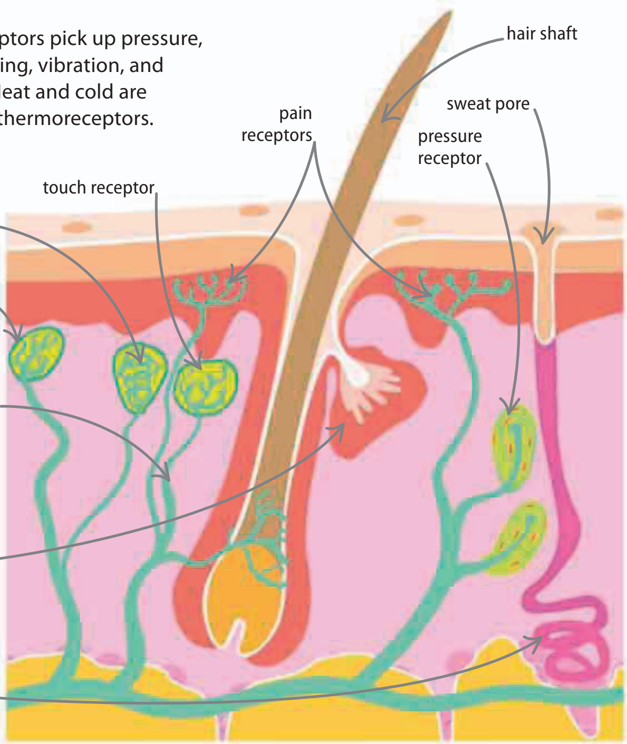
Smell and taste

Our senses of smell and taste both involve collecting chemicals and analysing them. The nose collects chemicals carried in the air. Inside the nose, scent chemicals dissolve in the mucus lining of the nasal cavity (which also helps clean the air). The chemicals are detected by hairlike nerve endings that send signals to the brain.The tongue detects similar chemicals in food.



▷ **Taste bud**
Taste buds are located on the tongue, gums, and throat. They have nerve endings covered in proteins that can detect specific chemicals associated with certain foods, such as sweet sugar or sour acids.

▷ **Skin**
Mechanoreceptors pick up pressure, touch, stretching, vibration, and sharp pains. Heat and cold are picked up by thermoreceptors.



Touch

The sense of touch relies on several types of receptors, mainly located in the skin, but also found in muscles, joints, and internal organs. There are about 50 touch receptors for every square inch of skin, although more sensitive body parts, such as the fingertips and tongue, have more, while the back has fewer.

REAL WORLD

Braille

Fingertips (touch), and not eyes (vision), are used to read Braille. Letters are represented by patterns of between one and five small bumps, or dots, arranged in a grid. Skilled Braille readers can read about 200 words per minute.